

Lezione del 22-5-2026

venerdì 22 maggio 2026 11:08

Consideriamo $(f(x))^{g(x)} : X \rightarrow \mathbb{R}$

ho zero se $f(x) > 0$ $\forall x \in X$

Ex

$$\left(1 + \frac{1}{x}\right)^x : X \rightarrow \mathbb{R}$$

Chi è X ?

$$f(x) = 1 + \frac{1}{x}$$

$$g(x) = x$$

R. di

Verificare come (non) si comporta, occorre che

$$1 + \frac{1}{x} > 0 \quad (\Rightarrow)$$

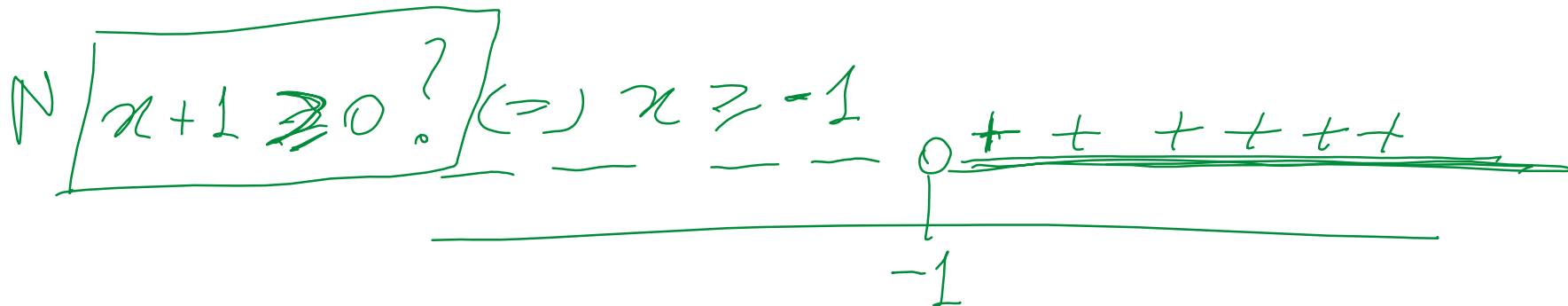
$$\boxed{\frac{x+1}{x} > 0}$$

Studiamo il segno di

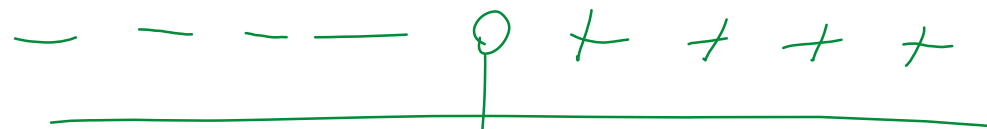
$$N : x+1$$

o he!

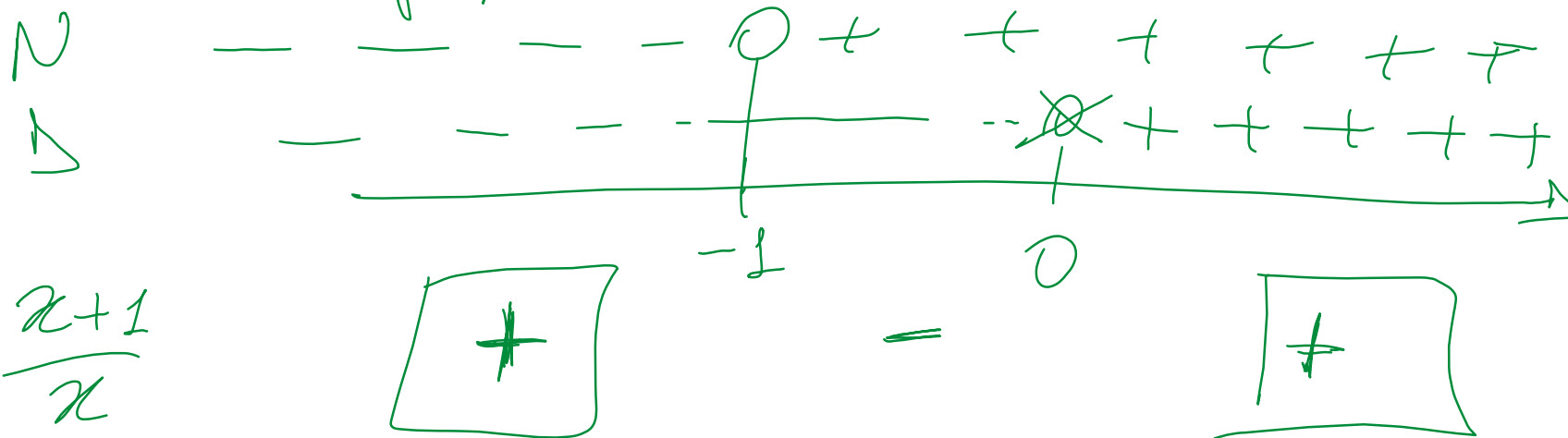
$$D : x$$



$$D : x$$



Unendo i pezzi



Bertando

$$\left(1 + \frac{1}{x}\right)^x :]-\infty, -1[\cup]0, \infty[\rightarrow$$

Caso due del

$\mathbb{Q} \setminus \mathbb{N}$

$$\lim_{x \rightarrow x_0^*} (f(x))^{g(x)}$$

$$1) \quad \lim_{x \rightarrow x_0^*} f(x) = l \in \mathbb{R} \quad l \neq 0 \quad \Rightarrow$$

$$\lim_{x \rightarrow x_0^*} f(x) = m \in \mathbb{R}$$

$$\lim_{x \rightarrow x_0^*} (f(x))^{g(x)} = l^m$$

Se non sono entrambi convergenti, allora

non si può dire nulla

convergenza puntuale $\sim$ γ -uniforme

$$(f_n)^{\log n}$$

nel seguente caso:

$$(K > 0 \Rightarrow K = A^{\log K})$$

$\forall x \in X$ si ha $(f(x) > 0 \quad \forall x \in X)$

$$f(x) = e^{\log f(x)}$$

allora

$$f(x) \leftarrow p_1, p_2, \dots, f(x)$$

$$(f(n)) = (e^{\log_e f(n)}) =$$

$$= e^{g(n) \log_e f(n)}$$

$$\lim_{n \rightarrow n_0^*} (f(n))^{g(n)} \Leftrightarrow \lim_{n \rightarrow n_0^*} e^{g(n) \log_e (f(n))}$$

Per questo, si calcola la parte

$$\lim_{n \rightarrow n_0^*} g(n) \log_e (f(n))$$

$$\lim_{x \rightarrow -\infty} \left(2 + \frac{1}{x}\right)^x$$

$$\begin{aligned} f(u) &= 2 + \frac{1}{x} \rightarrow 2 \\ \text{He } f(x) &= x \rightarrow -\infty \end{aligned}$$

$$\left(2 + \frac{1}{x}\right)^x = e^{x \cdot \log\left(2 + \frac{1}{x}\right)}$$

A parte $\lim_{x \rightarrow -\infty} x \log\left(2 + \frac{1}{x}\right) = (-\infty \cdot \log 2) = -\infty$

$$\lim_{x \rightarrow -\infty} e^{x \log\left(2 + \frac{1}{x}\right)} = \lim_{y \rightarrow -\infty} e^y = 0$$

e quindi $\lim_{x \rightarrow -\infty} \left(2 + \frac{1}{x}\right)^x = 0$

$x \rightarrow -\infty$ $x \rightarrow \infty$

F.l.

$\lim_{x \rightarrow x^*} (f(x))^{g(x)}$

$\boxed{1^\infty}$

SEMPRE
con NEPERO

F.l.

0^0

∞^0

Si prova che

$$\exists \lim_{x \rightarrow \pm \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$\left(1^\infty\right)$
F.l.

o ha il T nel limite delle costanti:

$$\text{se } \exists \lim_{x \rightarrow x_0^*} f(x) = \pm \infty$$

allora

$$\exists \lim_{x \rightarrow x_0^*} \left(1 + \frac{1}{f(x)} \right)^{f(x)} = e$$

EX

$$\lim_{x \rightarrow +\infty} \left(\frac{x^3 + 2^x + 1}{2^x + x^4} \right)^{\frac{1}{x}}$$

$$\sim \frac{x^3}{2^x} \rightarrow 0 \quad \dots \quad 1 \quad 1$$

$$\lim_{n \rightarrow \infty} \frac{2^n \left(\frac{1}{2^n} + 1 + \frac{1}{2^n} \right)}{2^n \left(1 + \frac{n^4}{2^n} \right)} = 1$$

$\rightarrow 0$

$$\Rightarrow F.I \quad 1^\infty$$

$$\left(\frac{n^3 + 2^n + 1}{2^n + n^4} \right)^{\frac{1}{n}} =$$

$$= \left(1 + \frac{n^3 + 2^n + 1}{2^n + n^4} \right)^{\frac{1}{n}} =$$

$, n$

$$= \left(1 + \frac{\cancel{2^n} - n^4 + n^3 + \cancel{2^n} + 1}{2^n + n^4} \right)^n =$$

$$= \left(1 + \frac{-n^4 + n^3 + 1}{2^n + n^4} \right)^n =$$

$$= \left(1 + \frac{1}{\frac{2^n + n^4}{-n^4 + n^3 + 1}} \right)^n =$$

$= f(n)$

$$2^n + n^4$$

$$-n^4 + n^3 + 1$$

2

$$= \left(1 + \frac{1}{\frac{2^n + n^4}{-n^4 + n^3 + 1}} \right)$$

$$\frac{-n^4 + n^3 + 1}{2^n + n^4} \cdot h$$

$$= \left[\left(1 + \frac{1}{\frac{2^n + n^4}{-n^4 + n^3 + 1}} \right) \frac{2^n + n^4}{-n^4 + n^3 + 1} \right]$$

$$\frac{h^n (-n^4 + n^3 + 1)}{2^n + n^4}$$

A parte

$$\lim_{n \rightarrow \infty} \frac{h^n (-n^4 + n^3 + 1)}{2^n (1 + \frac{n^4}{-n^4 + n^3 + 1})} = \lim_{n \rightarrow \infty} \left(\frac{h}{2} \right)^n \frac{-n^4}{1 + \frac{n^4}{-n^4 + n^3 + 1}}$$

- - - 2^x

$$= +\infty \cdot (-\infty) = -\infty$$

$$e^{-\infty} =$$

Portando

$$\lim_{x \rightarrow \infty} \left(\frac{x^3 + 2x + 1}{2^x + x^4} \right)^{\frac{1}{x}} = \left(e^{-\infty} \right) = 0$$

Ex

$$\lim_{x \rightarrow 0^+} x^{\log x}$$

$$\lim_{x \rightarrow 0^+} (x^{\log x})$$

$$x^{\log x} = \left(e^{\log x} \right)^{\log x}$$

$$= e^{\log x \cdot \log x}$$

$$\lim_{x \rightarrow 0^+} \log x \cdot \log x = (-\infty)(-\infty) = +\infty$$

Quindi

$$\lim_{x \rightarrow 0^+} x^{\log x} = \lim_{y \rightarrow +\infty} e^y = +\infty$$

Il limite (notevole)

$$\lim_{n \rightarrow \pm \infty} \left(1 + \frac{1}{n}\right)^n = e$$

parte con se' diversi risultati
anche omi notevoli:

$$\lim_{n \rightarrow 0} (1+n)^{\frac{1}{n}} = e$$

(Sostituisce
 $y = \frac{1}{n}$ per
avvicinare a

$$(1+n)^{\frac{1}{n}} \sim \left(1 + \frac{1}{y}\right)^y$$

$$\lim_{x \rightarrow 0} \frac{\log_e(1+x)}{x} = \log_e \left(\lim_{x \rightarrow 0} \frac{1+x}{1} \right) = \log_e e$$

In particolare

$$\lim_{x \rightarrow 0} \frac{\log_e(1+x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\log_e(1+x)}{x} = \lim_{x \rightarrow 0} \frac{1}{x} \cdot \log_e(1+x)$$

$$= \lim_{x \rightarrow 0} \log_e \left((1+x)^{\frac{1}{x}} \right) = \log_e e$$

$$k \log_e b = \log_e b^k$$

$$\lim_{n \rightarrow 0} \frac{A^n - 1}{n} = \log_e A \quad \left(\text{F.l. } \frac{0}{0} \right)$$

$$\lim_{n \rightarrow 0} \frac{e^n - 1}{n} = 1$$

$$\lim_{n \rightarrow 0} \frac{(1+n)^k - 1}{n} = k \quad \left(\frac{0}{0} \text{ F.l.} \right)$$

$$\text{In l. t. r. d. o.} \quad (1+n)^{\frac{1}{m}}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt[m]{1+x} - 1}{x} = \frac{1}{m}$$

grazie al T. sui limiti di FUNZIONI
COMPOSITE

Volgaro

h $\exists \lim_{x \rightarrow x_0^*} f(x) = 0$, allora

$$\exists \lim_{x \rightarrow x_0^*} (1 + f(x))^{1/f(x)} = e$$

$$\lim_{n \rightarrow \infty} (1 + \frac{1}{n}) = e$$

$$\exists \lim_{x \rightarrow x_0^*} \frac{\log_a(1 + f(x))}{f(x)} = \log_a e$$

$$\exists \lim_{x \rightarrow x_0^*} \frac{A^{f(x)} - 1}{f(x)} = \log_e A$$

$$\exists \lim_{x \rightarrow x_0^*} \frac{(1 + f(x))^K - 1}{f(x)} = K$$

Ex

$$\lim_{x \rightarrow 1} \frac{\log(2-x)}{x^2-1}$$

$$\left(\frac{\log(1) = 0}{1-1 \quad 0} \right)$$

$$\frac{\log(2-x)}{x^2-1} = \frac{\log(\cancel{1}-1+2-x)}{x^2-1} =$$

$$= \frac{\log(1+\cancel{1-x})}{x^2-1} \stackrel{\text{f.m.}}{\Rightarrow} \frac{1-x}{1-x} =$$

$$= \frac{\log(1+1-x)}{1-x} \cdot \frac{1-x}{x^2-1} \rightarrow -\frac{1}{2}$$

un'idea
nottevole

A parte

$$\lim_{n \rightarrow 1} \frac{1-n}{n^2-1}$$

$$\left(\frac{0}{0} \right)$$

→ -1/2

$$\frac{1-n}{n^2-1} = \frac{-1+n}{(n-1)(n+1)} \rightarrow \frac{-1}{2} \rightarrow -\frac{1}{2}$$

Pertanto

$$\lim_{n \rightarrow 1} \frac{\log(2-n)}{n^2-1} = \lim_{n \rightarrow 1} \frac{\log(1+1-n)}{1-n} \cdot \frac{-1}{n+1} = 1 \cdot (-1) = -1$$

Ex

$$\lim_{n \rightarrow +\infty} \frac{\frac{n+1}{n^2} - 1}{2^{-n}}$$

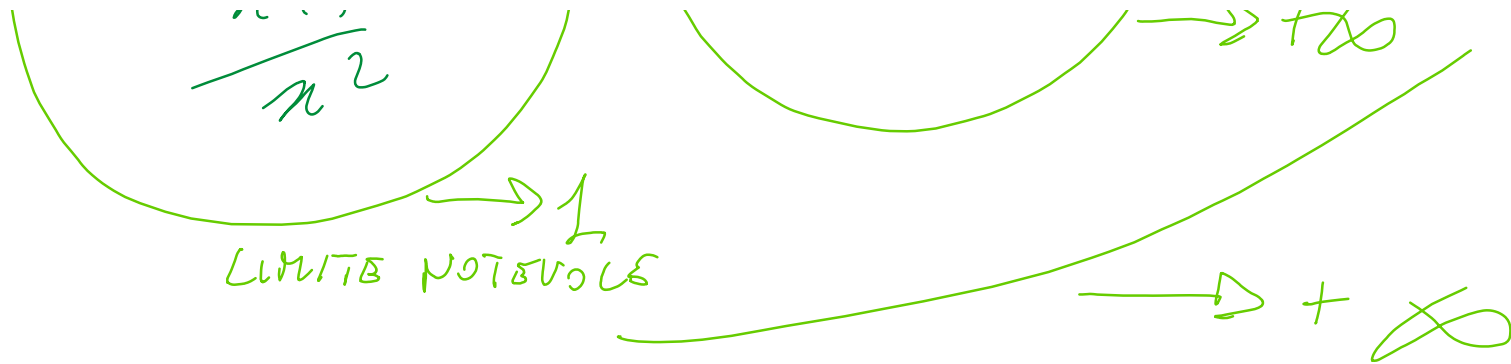
$\frac{n+1}{n^2} \rightarrow 0$ (circled in green)
 $2^{-n} \rightarrow 0$ (circled in green)

$$\begin{pmatrix} 1-1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\frac{\frac{n+1}{n^2} - 1}{2^{-n}} = \frac{\frac{n+1}{n^2} - 1}{\frac{n+1}{n^2}} \cdot \frac{n+1}{n^2} \cdot \frac{1}{2^{-n}}$$

$\frac{n+1}{n^2} = f(n) \rightarrow 0$ (circled in pink)

$$= \left(\frac{\frac{n+1}{n^2} - 1}{n+1} \right) \cdot \left(\frac{n+1}{n^2 2^{-n}} \right)$$



A parte $\lim_{n \rightarrow +\infty} \frac{n+1}{n^2 2^{-n}}$
 $+\infty = 0$

$$\frac{n+1}{n^2 2^{-n}} = 2^n \cdot \frac{n+1}{n^2} = \left(\frac{2^n}{n^2} \right) \cdot (n+1)$$

Hand-drawn graphs illustrating the limit notation. The first graph shows a curve increasing towards positive infinity, labeled "→ +∞". The second graph shows a curve increasing towards positive infinity, labeled "→ +∞". The third graph shows a curve increasing towards positive infinity, labeled "→ +∞".

$$\left(\frac{1}{2^{-n}} = \frac{1}{\frac{1}{2^n}} = 2^n \right)$$

Contenuto

$$\lim_{n \rightarrow +\infty} \frac{e^{\frac{n+1}{n^2}} - 1}{2^{-n}} = \lim_{n \rightarrow +\infty} \frac{e^{\frac{n+1}{n^2}} - 1}{\frac{n+1}{n^2}} \cdot \frac{2^n (n+1)}{n^2}$$

$$= 1 \cdot (+\infty) = +\infty$$

T (old content)

Sei $g, f, h : X \rightarrow \mathbb{R}$ tali che

$$g(x) \leq f(x) \leq h(x) \quad \forall x \in I_{x_0^*}, \text{ allora}$$

$$i) \quad \text{se } \exists \lim_{x \rightarrow x_0^*} g(x) = +\infty \Rightarrow \exists \lim_{x \rightarrow x_0^*} f(x) = +\infty$$

$$ii) \quad \text{se } \exists \lim_{x \rightarrow x_0^*} h(x) = -\infty \Rightarrow \exists \lim_{x \rightarrow x_0^*} f(x) = -\infty$$

iii) (T dei Cauchy)

$$\text{se } \exists \lim_{x \rightarrow x_0^*} g(x) = \lim_{x \rightarrow x_0^*} h(x) = l \in \mathbb{R}$$

allora

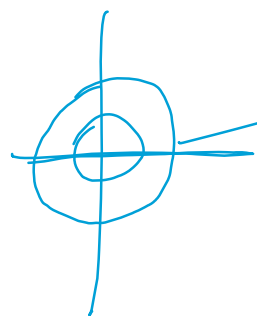
$$\exists \lim_{x \rightarrow x_0^*} f(x) = l \in \mathbb{R}$$

D1.17 Proviamo le iii) nel caso $x_0^* = x_0 \in \mathbb{R}$.

A CASA PROVARE i) e ii) nel caso $x_0^* = +\infty$

Occorre provare che:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 : \forall x \in X \cap]x_0 - \delta, x_0 + \delta[\setminus \{x_0\}$$



$$l - \varepsilon < f(x) < l + \varepsilon$$

Fissiamo in $\varepsilon > 0$.

Dal fatto che $\exists \lim_{x \rightarrow x_0} f(x) = l \quad \Rightarrow$

$$\exists \delta_1 > 0 : \forall x \in]x_0 - \delta_1, x_0 + \delta_1[\cap X \setminus \{x_0\}$$

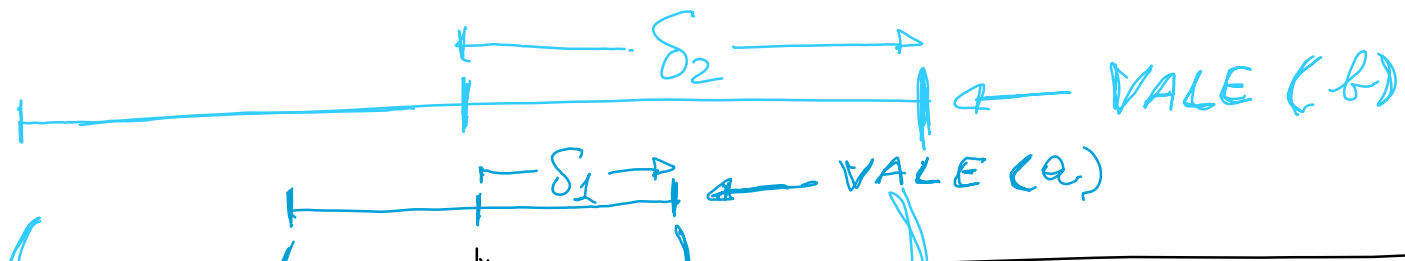
$$l - \varepsilon < g(x) < l + \varepsilon \quad (a)$$

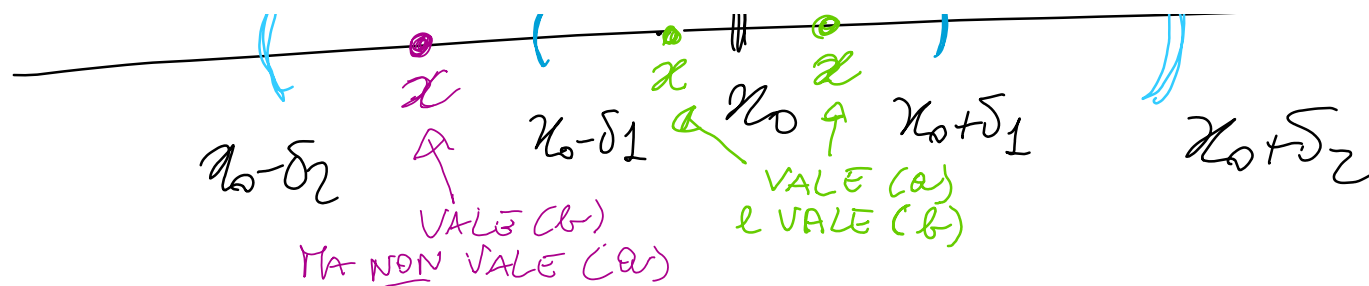
Dal fatto che $\exists \lim_{x \rightarrow x_0} h(x) = l$

$$\exists \delta_2 > 0 : \forall x \in]x_0 - \delta_2, x_0 + \delta_2[\cap X \setminus \{x_0\}$$

$$l - \varepsilon < h(x) < l + \varepsilon \quad (b)$$

GRAFICAMENTE ABBIAMO che (e per esempio: $\delta_2 > \delta_1$)





Scegliamo come δ delle Ter: $\delta = \min\{\delta_1, \delta_2\}$.

Per ogni $x \in]x_0 - \delta, x_0 + \delta[$, allora

(a) e (b) valgono contemporaneamente:

$$l - \varepsilon \leq f(x) \leq f(x) \leq h(x) \leq l + \varepsilon$$

$\uparrow$ (a) IPOTESI $\uparrow$ (b)

$$l - \varepsilon < l < l + \varepsilon$$

$$x - \epsilon < f(x) < x + \epsilon$$

ovvero

$$\exists \lim_{x \rightarrow x_0} f(x) = l$$

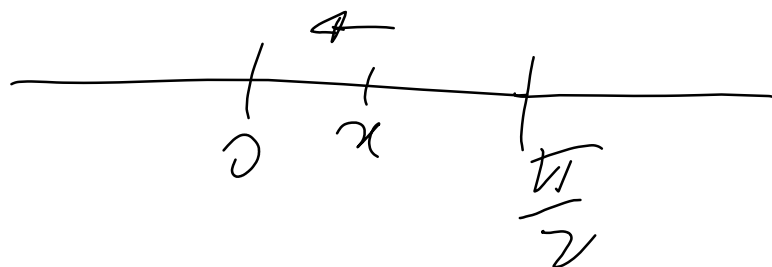
Applichiamo il T. dei Carabiniere per provare il seguente:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

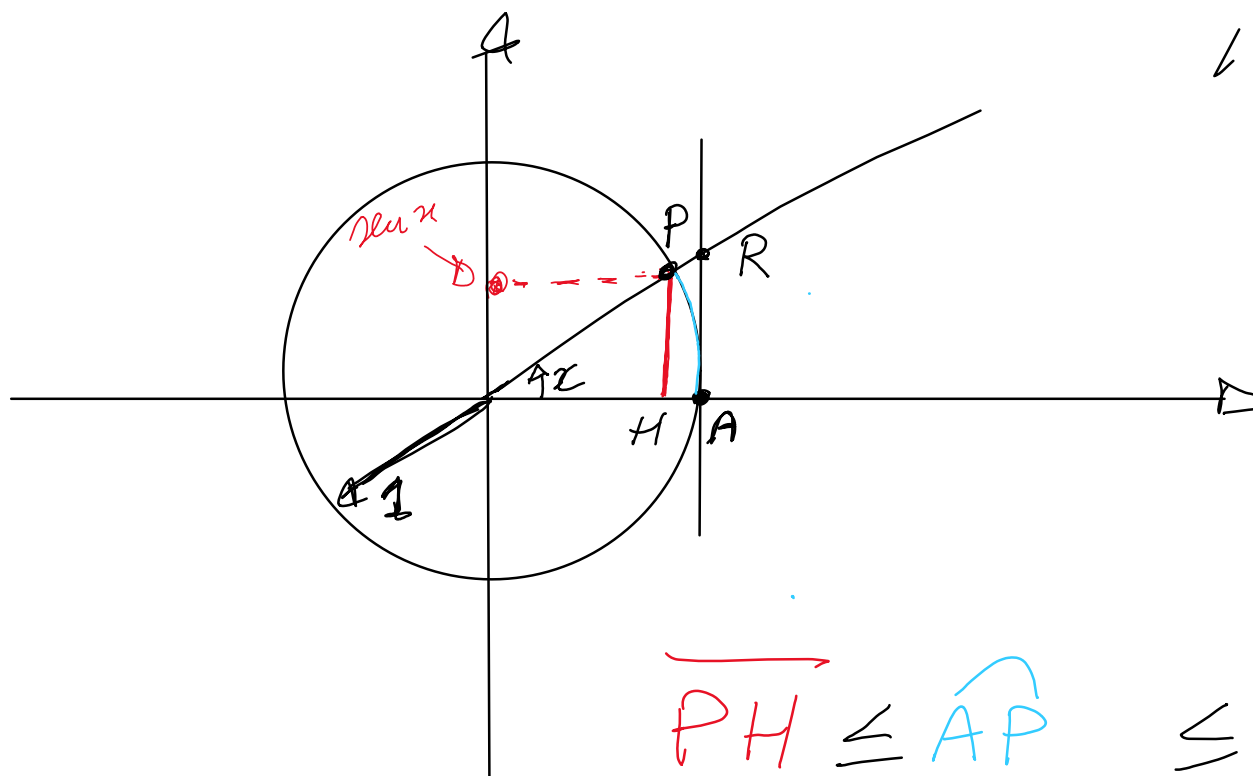
(F.I. $\frac{0}{0}$)

Per particolare possiamo provare:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

circa
 $x \rightarrow 0^+$
 x $\perp$
 $(\text{se } x \rightarrow 0^+)$
 $\text{sen } x$

und ihre $x \in [0, \pi/2]$ in RADIAN!



$$\overline{PH} \leq \widehat{AP} \leq \overline{AR}$$

$$\overline{PH} = \operatorname{sen} x \quad (x > 0)$$

$$\overline{AP} = x$$

$$\overline{AR} = \operatorname{tg} x \quad (x > 0)$$



$$(0 <) \operatorname{sen} x \leq x \leq \operatorname{tg} x$$

Dividendo per $\operatorname{sen} x > 0 \quad (x > 0)$

$$1 \leq \frac{x}{\operatorname{sen} x} \leq \frac{\operatorname{tg} x}{\operatorname{sen} x} = \frac{\operatorname{sen} x}{\cos x} \cdot \frac{1}{\operatorname{sen} x} = \frac{1}{\cos x}$$

Quindi

$$1 \leq \frac{x}{\operatorname{sen} x} \leq \frac{1}{\cos x} \quad \forall x \in]0, \frac{\pi}{2}]$$

$\lim_{x \rightarrow 0} \frac{\sin x}{x}$

Passando ai reciproci:

$$(1) \quad \cos x \leq \frac{\sin x}{x} \leq 1 \quad \forall x \in]0, \frac{\pi}{2}]$$

$$\begin{array}{c} \downarrow x \rightarrow 0 \\ \triangle \\ 1 \end{array}$$

$$\begin{array}{c} \downarrow x \rightarrow 0 \\ 0 \\ 1 \end{array}$$

$$\begin{array}{c} \downarrow \\ \triangle \\ 1 \end{array} \quad \leftarrow \text{T. dei Cordoni}$$

Potendo

$$\exists \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$$

1. $x \rightarrow -x$

$$\text{Se } x \in [-\frac{\pi}{2}, 0[\Rightarrow -x \in]0, \frac{\pi}{2}]$$

allora posto $y = -x \in]0, \frac{\pi}{2}]$

valle le (1) con y al posto di x

cioè

$$\cos y \leq \frac{\sin y}{y} \leq 1$$

per tanto

$$\cos(-x) \leq \frac{\sin(-x)}{-x} \leq 1 \quad \forall x \in]-\frac{\pi}{2}, 0]$$

$$\begin{array}{ccc} \downarrow & & \downarrow \quad \downarrow \\ \cos x & \leq & \frac{-\sin x}{-x} \leq 1 \end{array}$$

$$\cos x \leq \frac{\sin x}{x} \leq 1$$

$$\forall x \in]-\frac{\pi}{2}, 0]$$

$\Rightarrow$ per il T. dei coseno:

$$\exists \lim_{x \rightarrow 0^-} \frac{\sin x}{x} = \underline{1}$$

Per case

$$\lim_{n \rightarrow \infty} \frac{\sqrt[3]{n} + 1}{n-3} ; \lim_{n \rightarrow \infty} \frac{n + (-1)^{n+1}}{2n + \cos(n\pi)}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n^2 + 2n + 2}{n^2 + 1} \right)^{n^2 - 1} ; \lim_{n \rightarrow \infty} \sqrt[n]{n^3 + 3n + 2}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n^2 + 3^n + 2^n} ; \lim_{x \rightarrow 1} \frac{e^{x^3 - 1} - 1}{x - 1}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{5 - x^2} - (x + 1)}{x^2 - 1} ; \lim_{x \rightarrow 2} \frac{e^x - e^2}{x^2 - 4}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + x^2} - 1}{3x^2 + 1 - e^x} ; \lim_{x \rightarrow 0} (1 + 3x^2)^{\frac{1}{\ln^2 x}}$$

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 3x} - x}{\sqrt{x^2 - 1} - x}$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x}$$